CHAPTER 5

INTRODUCTION TO C++ PROGRAMMING

CHAPTER ESSENCE AT A GLANCE:
Master the foundational syntax and build structure of C++ programming encompassing lexical tokens, structured preprocessor directives, input-output stream chaining, and executable build toolchains.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Lexical Token Classification:** C++ source code decomposes into fundamental lexical tokens: reserved keywords, user-defined identifiers, immutable literal constants, punctuators, and symbolic operators.

• **Identifier Construction Rules:** Valid identifiers begin with an alphabet or underscore, contain alphanumeric characters, exclude special symbols and spaces, and are strictly case-sensitive.

• **Preprocessor and Namespace Hierarchy:** The #include directive instructs the preprocessor to inject header files, while 'using namespace std;' exposes standard library symbols into global scope.

• **Stream Chaining Mechanics:** The insertion operator cascades formatted data to standard output, while the extraction operator parses incoming typed tokens sequentially from standard input.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{cout} \ll \text{"Result: "} \ll \text{val} \ll \text{endl;} \quad \text{(Stream Insertion)}$$

Cascading Standard Output Stream

$$\text{cin} \gg \text{var1} \gg \text{var2;} \quad \text{(Stream Extraction Chaining)}$$

Cascading Standard Input Stream

$$\text{Source Code (.cpp)} \xrightarrow{\text{Compiler}} \text{Object (.o)} \xrightarrow{\text{Linker}} \text{Executable}$$

C++ Compilation and Build Pipeline

$$\text{Escape Sequences} \in \{\backslash\text{n (Newline)}, \backslash\text{t (Tab)}, \backslash\backslash \text{ (Backslash)}\}$$

Non-Graphic Literal Formatting Characters

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In modern standard C++, always include header files without the legacy .h extension: use #include <iostream> instead of <iostream.h>.

• **Alert:** Do not confuse stream insertion << (used with cout) with stream extraction >> (used with cin).

• **Alert:** Reserved keywords cannot be used as variable identifiers (for instance, declaring 'int float;' is illegal).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is a Token in C++? Name the five types of tokens supported in C++. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Token Definition:** The smallest individual lexical unit in a C++ program that possesses meaning to the compiler. **[1 Mark]**

• **Five Types of Tokens:** (1) Keywords, (2) Identifiers, (3) Literals (Constants), (4) Operators, and (5) Punctuators. **[1 Mark]**

2. State any four rules for naming identifiers in C++. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Rule 1 & 2:** An identifier must begin with an alphabet letter (a-z, A-Z) or an underscore; it can contain letters, digits, and underscores. **[1 Mark]**

• **Rule 3 & 4:** It must not contain whitespace or special symbols (such as @, $, %); it cannot be a reserved C++ keyword; and it is case-sensitive (e.g. Count and count are distinct). **[1 Mark]**

3. Explain the structure of a standard C++ program with an annotated code outline. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Preprocessor Directives & Namespace:** #include <iostream> directives instruct compiler to include input-output declarations; using namespace std; imports standard names. **[1 Mark]**

• **Main Function:** int main() serves as the mandatory execution entry point of every C++ program enclosed within curly braces {}. **[1 Mark]**

• **Body & Return Statement:** Contains variable declarations, executable algorithmic statements, and terminates with 'return 0;' signaling successful exit. **[1 Mark]**

4. Explain the cascading of I-O operators in C++ with suitable syntax examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Concept:** Cascading refers to using the insertion operator << or extraction operator >> multiple times consecutively in a single C++ statement. **[1 Mark]**

• **Cascading Output:** cout << 'Roll: ' << roll << ' Mark: ' << mark << endl; sends multiple strings and variables sequentially to screen. **[1 Mark]**

• **Cascading Input:** cin >> a >> b >> c; reads three continuous values from keyboard into variables in a single line. **[1 Mark]**

5. Identify which of the following are valid and invalid C++ identifiers, giving reasons: (a) 2ndNumber, (b) total-sum, (c) default, (d) studentName. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) 2ndNumber:** Invalid. Identifiers cannot begin with a numeric digit. **[0.5 Mark]**

• **(b) total-sum:** Invalid. The hyphen is an arithmetic minus operator and cannot be used in variable identifiers. **[1 Mark]**

• **(c) default:** Invalid. 'default' is a reserved keyword in C++ used in switch statements. **[0.5 Mark]**

• **(d) studentName:** Valid. Follows camelCase conventions with alphabet start and no illegal characters. **[1 Mark]**

6. Write a complete C++ program to input length and breadth of a rectangle and display its area and perimeter. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Header & Main Setup:** #include <iostream>vsing namespace std;nint main() { float length, breadth, area, perimeter; **[1 Mark]**

• **Input & Computation:** cout << 'Enter length and breadth: '; cin >> length >> breadth;narea = length * breadth;nperimeter = 2 * (length + breadth); **[1 Mark]**

• **Output & Termination:** cout << 'Area: ' << area << endl << 'Perimeter: ' << perimeter << endl; return 0; } Boxed Result: Clean compilation with accurate arithmetic. **[1 Mark]**

7. Explain escape sequences in C++ and state the output of the statement: cout << 'HellonWorldtDone';
[Score: 3 Marks]

Answer & Key Valuation Steps:

• **Escape Sequence Concept:** Non-graphic character combinations starting with a backslash that trigger special formatting actions during console output. **[1 Mark]**

• **Components:** n inserts a newline; t inserts a horizontal tab jump; \\ displays a literal backslash. **[1 Mark]**

• **Output Evaluation & Boxed Result:** 'Hello' on first line, followed by 'World', a tab space, and 'Done' on the second line. Boxed Result: Evaluates cleanly across two formatted output lines. **[1 Mark]**